

```
import cv2
from IPython.display import Image, display

# Using the actual filename available in your environment
filename = 'upscalemedia-transformed.jpeg'
img = cv2.imread(filename)

if img is None:
    print(f"Error: Could not load image '{filename}'. Please ensure the fi:
else:
    height, width = img.shape[:2]
    scale_factor = 2.0
    new_width_large = int(width * scale_factor)
    new_height_large = int(height * scale_factor)

    img_bigger = cv2.resize(
        img,
        (new_width_large, new_height_large),
        interpolation=cv2.INTER_CUBIC
    )

    new_width_small = int(width * 0.5)
    new_height_small = int(height * 0.5)

    img_smaller = cv2.resize(
        img,
        (new_width_small, new_height_small),
        interpolation=cv2.INTER_AREA
    )

    cv2.imwrite('bigger_image1.jpg', img_bigger)
    cv2.imwrite('smaller_image1.jpg', img_smaller)
    print("Images processed and saved: 'bigger_image1.jpg' and 'smaller_im:

    # Displaying the results
    print("\n--- Visual Results ---")
    print("Original Image:")
    display(Image(filename=filename, width=400))

    print("\nUpscaled (2x):")
    display(Image(filename='bigger_image1.jpg', width=400))

    print("\nDownscaled (0.5x):")
    display(Image(filename='smaller_image1.jpg', width=200))
```

Images processed and saved: 'bigger_image1.jpg' and 'smaller_image1.jpg'

--- Visual Results ---

Original Image:



Upscaled (2x):



Downscaled (0.5x):

